

University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Term Final Examination, Spring 2025
Course Title: Database Management System
Course Code: CSE0612215

Marks: 50

Time: 3(three) hours

(Answer all questions)

Q.No.

Questions

Marks

1

Consider the ER diagram of a bank account database for a customer

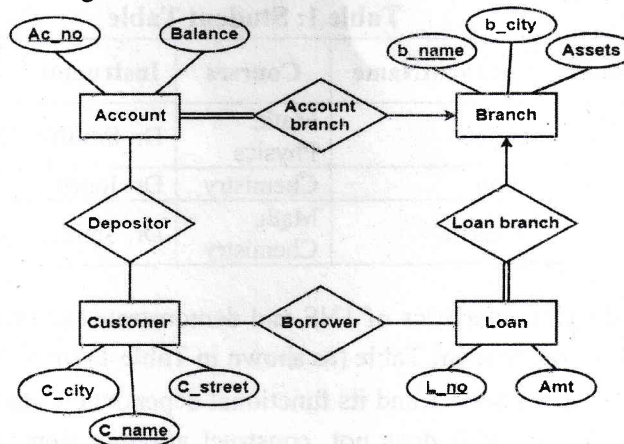


Figure 1: Customer account database

- Write the relational schema to represent the ER diagram of Customer account database shown in Figure 1. [04]
- Illustrate the type of mapping cardinality of the Customer account database from figure 1. [06]

2.

Consider the following relational schema to write the relational algebra:

Student(StudentID, Name, DeptID)

Department(DeptID, DeptName, Building)

Course(CourseID, Title, DeptID)

Enrolled(StudentID, CourseID)

- Find all combinations of students and courses, and then filter only those where the student belongs to the same department offering the course. [03]
- List all students name along with the names of the courses they have enrolled in. [03]
- Explain the difference between the COUNT() and SUM() aggregate functions, including an example query for each. [04]

3.

Consider the following relational schema:

Book(BookID PK, Title, ISBN, Publisher, PublicationYear, Price)

Author(AuthorID PK, Name, Bio)

Member(MemberID PK, Name, Address, Phone, Email, MembershipDate)

Copy(CopyID PK, BookID FK, Condition, AvailabilityStatus)

Loan(LoanID PK, CopyID FK, MemberID FK, LoanDate, DueDate, ReturnDate, FineAmount)

Category(CategoryID PK, CategoryName)

BookAuthor(BookID PK, AuthorID PK)

- a) Design the ER diagram for this schema. [07]
- b) Demonstrate Specialization and Generalization with an example. [03]

4.

Consider the following Table 1 of the University Database:

Table 1: Student Table

StudentID	StudentName	Courses	Instructor	Instructor Office
101	Alice	Math, Physics	Dr. Smith	Room 101
102	Bob	Chemistry	Dr. Jones	Room 102
103	Charlie	Math, Chemistry	Dr. Smith	Room 101

- a) Describe the characteristics of 1NF and demonstrate the process of converting the data from the Student Table (as shown in Table 1) into 1NF. [04]
- b) Analyze the given table 1 and its functional dependencies to determine whether it satisfies BCNF. If it does not, construct an equivalent decomposition that meets BCNF. [06]

5.

A manager needs to generate a report of employees who meet the following criteria:

Belong to the 'Sales' department
 Have a salary between 50,000 and 70,000
 Were hired after January 1, 2022

The report should display the EmployeeID, FirstName, LastName, Salary, and HireDate of the selected employees. The results must be sorted by HireDate in ascending order.

- a) Write an SQL query to retrieve this information from the Employee table. [05]
- b) Now, the manager wants to update the salary of all employees in the 'Sales' department by increasing it by 10%. Write the SQL command to perform this update operation. [05]